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1. AXIOMS FOR A GROUP

To define a group, we begin with a set G equipped with a *composition law*

$$\cdot : G \times G \longrightarrow G, \quad (g, h) \longmapsto g \cdot h, \quad (1.1)$$

and a distinguished element $e \in G$, called the *identity*. Thus the basic data are written $(G, \cdot, e)$. We often suppress the dot and write gh in place of $g \cdot h$.

Definition 1.1 (Group). The data $(G, \cdot, e)$ form a *group* if the following three conditions hold.

(1) **Associativity.** For all $g, h, k \in G$,

$$g \cdot (h \cdot k) = (g \cdot h) \cdot k. \quad (1.2)$$

(2) **Identity.** For every $g \in G$,

$$e \cdot g = g \cdot e = g. \quad (1.3)$$

(3) **Inverses.** For every $g \in G$, there exists $h \in G$ such that

$$h \cdot g = e \quad \text{and} \quad g \cdot h = e. \quad (1.4)$$

The first equality says that h is a *left inverse* of g ; the second says that it is a *right inverse*. So inverse really means ‘double sided’ inverse.

Associativity does *not* say that

$$g \cdot h = h \cdot g. \quad (1.5)$$

An operation satisfying this latter identity for all elements is called *commutative*.

Definition 1.2 (Abelian group). A group whose operation is commutative is called *abelian*.

It is also useful to observe the following simple result.

Lemma 1.1 (Uniqueness of inverses). *The element h in axiom (3) is unique.*

Proof. Suppose h_1 and h_2 both satisfy axiom (3) for g . Then

$$h_1 g = h_2 g = e. \quad (1.6)$$

Multiplying on the right by h_2 and using associativity gives

$$(h_1 g) h_2 = (h_2 g) h_2, \quad \text{hence} \quad h_1 (g h_2) = h_2 (g h_2). \quad (1.7)$$

Since $gh_2 = e$, it follows that $h_1 = h_2$. We write the unique inverse of g as g^{-1} . □

2. ASSOCIATIVITY, CYCLIC GROUPS, AND ORDER

The associativity axiom allows us to unambiguously refer to

$$g \cdot (h \cdot k) \text{ and } (g \cdot h) \cdot k \text{ as} \tag{2.1}$$

$$g \cdot h \cdot k. \tag{2.2}$$

An easy induction then allows us to define, given $g_1, \dots, g_n \in G$ an element

$$g_1 g_2 \cdots g_n \tag{2.3}$$

by removing parentheses in any order. Note again that we cannot change the order of the factors in the product (unless the group is abelian).

For $x \in G$ and a positive integer n , we also write

$$x^n = \underbrace{x \cdot x \cdots x}_{n \text{ factors}}. \tag{2.4}$$

If G is finite as a set, its cardinality $|G|$ is also called the *order* of G . If G is infinite, we write $|G| = \infty$ and say that G has infinite order.

3. EXAMPLES

Example 3.1 (The integers under addition). The triple

$$(\mathbb{Z}, +, 0) \tag{3.1}$$

is a group. Its composition law is

$$\mathbb{Z} \times \mathbb{Z} \longrightarrow \mathbb{Z}, \quad (m, n) \longmapsto m + n. \tag{3.2}$$

The identity element is 0, and the inverse of m is $-m$.

Example 3.2 (The integers under multiplication: a non-example). The triple

$$(\mathbb{Z}, \times, 1) \tag{3.3}$$

is *not* a group, although it is a semigroup. Multiplication is associative and 1 is an identity, so axioms **(1)** and **(2)** hold. Axiom **(3)** fails because most integers do not have multiplicative inverses in $\mathbb{Z}$.

Example 3.3 (Integers modulo n). Let n be a positive integer. The set of residue classes modulo n is

$$\mathbb{Z}/n\mathbb{Z} = \{\bar{0}, \bar{1}, \bar{2}, \dots, \overline{n-1}\}. \quad (3.4)$$

For example,

$$\begin{aligned} \bar{0} &= \{0, \pm n, \pm 2n, \pm 3n, \dots\}, \\ \bar{1} &= \{1, 1 \pm n, 1 \pm 2n, 1 \pm 3n, \dots\}. \end{aligned}$$

Every integer can be written $m = an + r$ with $0 \leq r < n$, and then $\overline{m} = \bar{r}$. Addition is defined by

$$\overline{m} + \bar{\ell} = \overline{m + \ell}. \quad (3.5)$$

Thus

$$(\mathbb{Z}/n\mathbb{Z}, +, \bar{0}) \quad (3.6)$$

is a group, called the *cyclic group of order n* .

Definition 3.1 (Cyclic group). A group G is *cyclic* if there is an element $x \in G$ such that

$$G = \{x^k : k \in \mathbb{Z}\}. \quad (3.7)$$

In this case, x is called a *generator* of G .

In the preceding examples that are groups, the group operation is both associative and commutative.

Example 3.4 (The general linear group). The set

$$\mathrm{GL}_n(\mathbb{R}) = \{A : A \text{ is an invertible } n \times n \text{ matrix with entries in } \mathbb{R}\} \quad (3.8)$$

is a group under matrix multiplication. Its identity is the identity matrix

$$I_n = \begin{pmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{pmatrix}. \quad (3.9)$$

Thus the group is written $(\mathrm{GL}_n(\mathbb{R}), \cdot, I_n)$, and the composition law sends (A, B) to AB .

Example 3.5 (The permutation group). The permutation group is

$$S_n = \{\text{all permutations of } \{1, 2, \dots, n\}\}. \quad (3.10)$$

Its group operation is composition of permutations.